



THE UNIVERSITY *of* EDINBURGH
School of Engineering

Engineering Portfolio

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I love making things, but I'm bad at taking pictures of my work.

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1 European Space Agency (ESA) - CEC 25

In April 2025, I was selected to participate in the European Space Agency's concurrent engineering challenge CEC25. The challenge requires a team of 30 students to assess the mission requirements and feasibility in the span of 5 days. I was selected as a Configuration Engineer, working on a team of three responsible for the entire physical design of the spacecraft. Our job was to figure out how every single component fit together in SOLIDWORKS. This was a constant puzzle of placing thrusters, casing electronics, and designing the deployment of the solar panels and antennas, all while considering things like thermal impacts and the needs of other teams. A huge part of our work was providing the AOCS and Propulsion teams with the numbers they needed—specifically, the spacecraft's Moment of Inertia (MoI) and Center of Gravity (CoG), which we calculated and updated as the design evolved.

Mission Objectives

- **Primary Objective 1:** Analyze the comet's chemical composition to gain insight into the formation of the Solar System and the origin of life.
- **Primary Objective 2:** Characterize the comet's mass distribution and shape to inform potential future planetary defense strategies.
- **Secondary Objective:** Assess the comet's resource potential for commercial applications, such as in-space fuel depots for future missions.

Preliminary Mission Phases

Phase	Location	Notes
LAUNCH	EARTH	Launch with european launcher
LEOP		Sun acquisition, spacecraft comissioning
L2 TRANSFER	DEEP SPACE	Transfer to L2 point
L2 ORBIT INSERTION	L2	Entering L2 parking orbit
WAITING FOR TARGET		Waiting for final choice of target
TARGET TRANSFER	DEEP SPACE	Transfer to selected comet
TARGET ORBIT INSERTION	COMET	Approach to comet
COMET MAPPING		Comet broad survey, choosing the landing site
TRIAL LANDING		Testing the landing capabilities, landing site detailed survey
COMET TOUCHDOWN		Landing on selected site
COLLECTING SAMPLES		Samples collection
IN-SITU SCIENCE		OPTIONAL – Further analysis from the surface
COMET LIFTOFF		Launch from the comet surface and return to its orbit
REPEAT OF SAMPLING CYCLE		OPTIONAL – Repeat of landing cycle for new selected site
EARTH TRANSFER	DEEP SPACE	Transfer back to Earth
CAPSULE SEPARATION	EARTH	Capsule release from the spacecraft on hyperbolic orbit
SPACECRAFT DIVERSION MANOUVERS		Changing the orbit of the spacecraft after capsule delivery
CAPSULE RE-ENTRY		Capsule enters the atmosphere
SAMPLE TOUCHDOWN		Capsule landing and sample retrieval by the ground team
SUN ORBIT		Graveyard orbit or.... mission extension?

Figure 1: Mission Phases Overview

My Contributions

- Throughout the challenge, we designed and assembled a total of 455 components for the satellite, integrating work from all subsystems into a master SOLIDWORKS model.

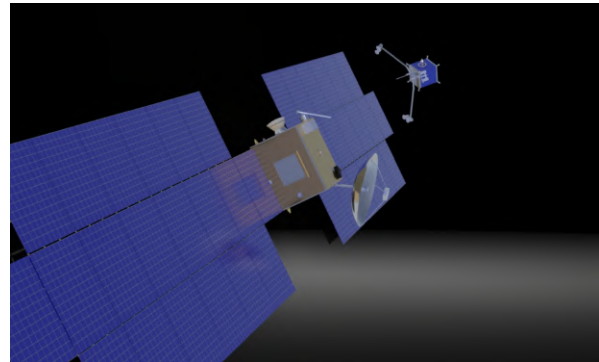
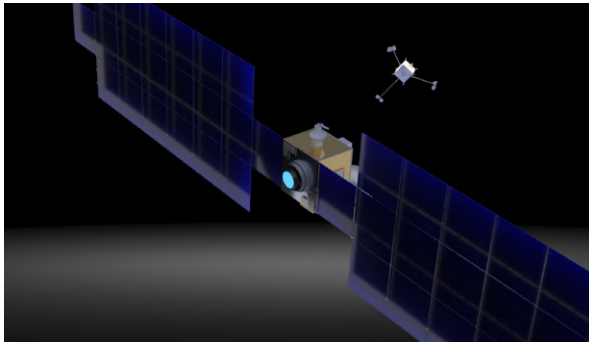
- The main challenge was the ever-changing requirements from other teams. With the concurrent design process and the tight deadline, components were constantly being updated, requiring constant reconfiguration.
- To manage this, our SOLIDWORKS design had to be smart and utilize variables. This allowed us to quickly adapt the layout to the evolving needs of the other sub-teams.
- Designing the deployment and orientation of large-area solar panels to meet power and thermal requirements.
- Packaging the lander module, its mini-lab, and the robotic docking/sampling arm within the main bus.
- Strategically placing AOCS thrusters to ensure effective attitude control without plume impingement on sensitive instruments.
- Produce CAD drawings of the satellite.
- **System-Level Analysis:** Performed early and continuous calculations for the spacecraft's total mass, center of gravity (CoG), and moment of inertia (Mol), providing critical data to the AOCS and Propulsion teams for their trajectory and control analysis.

(Note: The SOLIDWORKS model is intentionally blocky and rough. Our responsibility in this feasibility study was the correct volume and mass allocation for the preliminary design stage, not creating highly detailed parts.)

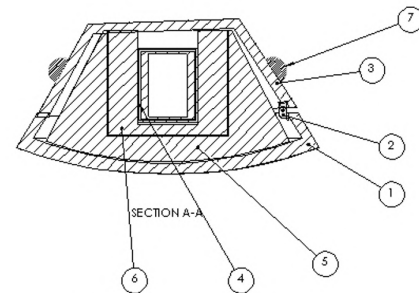
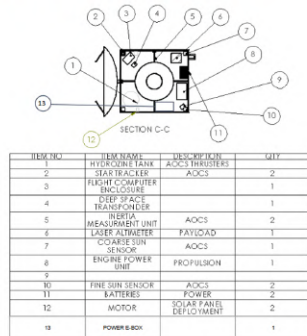
Skills Demonstrated

- **Engineering:** Concurrent Design, System Integration, Configuration Management
- **Tools:** SOLIDWORKS (Parametric Design, Assemblies), ESA COMET
- **Analysis:** Mass Properties (CoG, Mol), Kinematics, Clash Detection
- **Collaboration:** Working with AOCS, Propulsion, Structures, & Payload Subsystems
- **Professional:** Problem-Solving Under Pressure, Adapting to Evolving Designs

Figure 2: ESA CEC 25 Project Gallery



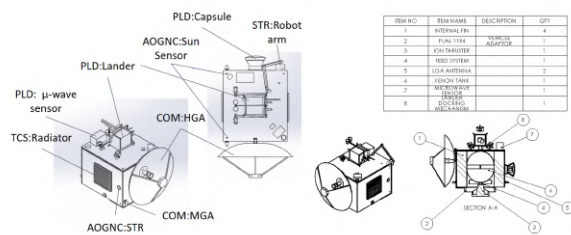
Orbiter layout-Top section view



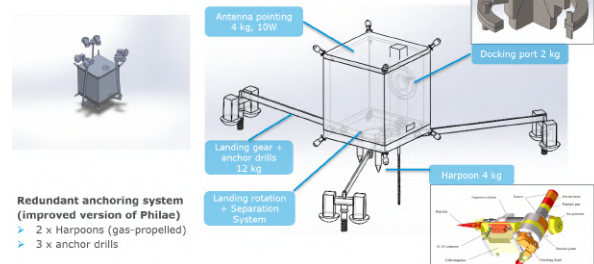
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Payload capsule	Ablative heat shield	1
2	Capsule hinge	Servo actuated hinge	4
3	Payload capsule_2	Aft. heat shield	1
4	Vacuum sealed sample capsule	Vacuum flask	1
5	External insulation	Aerogel	1
6	Internal insulation	MLI	1
7	Recovery system	Re-entry parachute	2

2025

Orbiter layout

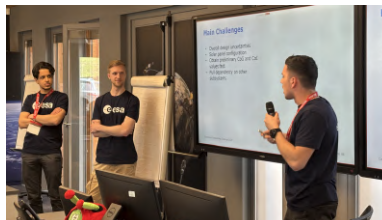
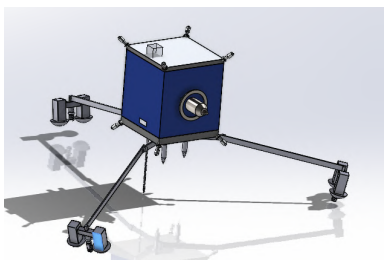


Final Baseline Design: Lander



Redundant anchoring system (improved version of Philae)

- 2 x Harpoons (gas-propelled)
- 3 x anchor drills



2 Endeavour Rocketry

2.1 Technical Lead - MACH 25

Context As Technical Lead for the academic year 24-25, I co-managed Endeavour's 30-person sounding rocket division. Our objective was to design, build, and launch the ADA-II rocket for the Mach 25 competition (3km apogee category). Our design philosophy prioritized rapid, low-cost iteration by scaling down complex designs and adopting robust, proven practices from the hobbyist rocketry community.

Mission Statement "The sounding rocket ADA-II, is to carry a deployable payload to an apogee of 3km, and safely recover the rocket in a flight-ready condition. The payload will feature a CanSat rover capable of rudimentary sample collection, vibration analysis, and flight stage prediction using machine learning algorithms."

My Contributions (as Technical Lead)

- **Project & Team Management:** Co-managed team logistics, including budgeting, procurement, and scheduling. I led technical meetings with subsystem leads to ensure project milestones were met and design integration was seamless.
- **Technical Strategy & Process:** Implemented the highly iterative ASIMOV design approach (later), ensuring all critical components and manufacturing techniques were flight-tested and validated prior to integration.
- **Safety, Compliance & Risk Management:** Authored and secured approval for all technical risk assessments, including those for advanced composites manufacturing. Obtained an explosives license and certified storage for rocket motors and black powder.
- **Mentorship Training:** Developed and delivered workshops to teach junior members the fundamentals of rocketry, CAD, and design for manufacturing.
- **External Relations Funding:** In response to university-wide budget cuts, I took the lead on securing external sponsorship to ensure the project remained funded.

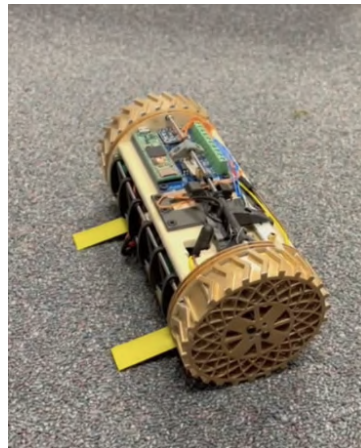
Table 1: Main Technical Specifications of ADA-II

Parameter	Value
Length	171 cm
Width	10.2 cm
Wet Mass	9.778 Kg
Dry Mass (w/ Payload)	8.170 Kg
Predicted Apogee	2900m (w/ 5° rail tilt)
Predicted Top Speed	293 m/s (Mach 0.871)
Rocket Motor	CTI L645-GN • Thrust (Avg) - 644 N • Burn Time - 5.33 s • Impulse - 3434 Ns
Peak Acceleration	72 m/s ² (7.34G)

Project Outcome While the final launch was canceled due to hazardous weather closing the competition window, the rocket was fully flight-ready and validated. Our team's efforts were recognized with the Best Outreach Award. The vehicle is scheduled for launch at the next available opportunity.

Skills Demonstrated Team Leadership, Project Management, Systems Engineering, Risk Assessment & Management, Budgeting & Procurement, Technical Training & Mentorship, Design Review & Drawing Verification.

Figure 3: MACH 25 Project Gallery



2.2 ASIMOV: Our Rocket Testing Program

Context We always had issues with not testing our components enough before a competition, so the ASIMOV program was created to tackle that problem head-on. The idea was simple: build a series of small, cheap rockets to test different subsystems on each flight.

Contributions

- **Iterative Process:** The plan was to start with hobby-style rockets and have each new version get closer to our competition rocket's final form, integrating more systems each time. This allowed us to fail small and learn fast.
- **Handled Logistics and Budget:** I supervised the program and handled all the logistics. We performed 6 different launches in one academic year, and I made sure each rocket's manufacturing cost stayed under £100.
- **Built Up Team Experience:** The best part of the program was seeing multiple team members get real, hands-on rocketry experience, from the workshop to the launch

field. We established a good routine with our launch procedures and, obviously, had fun doing it.

Figure 4: ASIMOV Program Launches



2.3 Structures Lead - FULMAIR

Context I was appointed Structures Lead for the FULMAIR rocket project, initially designed for the 2024 Spaceport America Cup. As a second-year engineer leading a team for the first time, I dedicated my summer to self-study in rocketry, FEA (ANSYS), and advanced CAD, developing training materials to onboard my team. Although university funding and supply chain issues prevented us from competing, the project's core design philosophy allowed it to pivot and continue as a testbed for our society's first liquid bi-propellant engine.

My Contributions (as Structures Lead)

- **Team Leadership & Mentorship:** Led and mentored a structures team of 8 junior members, providing training in SOLIDWORKS, ANSYS for finite element analysis, and design for manufacturing principles.
- **Design Philosophy & Architecture:** Established a core design philosophy centered on simplicity, accessibility, and modularity, enabling assembly and disassembly of the entire airframe with a single screwdriver.
- **Modular System Innovation:**

- Designed a novel universal bulkhead system with interlocking male/female components, enabling rapid swapping of the rocket's primary modules (payload, recovery, propulsion).
- Engineered the society's first removable fin can that also functioned as the motor's load-bearing thrust ring. This design successfully isolated the aerodynamic fins from primary structural loads.
- Designed and prototyped an airbrake system.
- **Design for Manufacturing (DFM):** Produced and verified a complete set of engineering drawings for all structural components, ensuring all parts were designed with machine constraints in mind. I was also responsible for material selection and composites manufacturing processes.
- **Validation Reporting:** Led technical design reviews with university faculty, workshop technicians, and UKRA safety officers. Authored the structural design and analysis sections for the Spaceport America submission report.

Key Lessons Learned

- **Redundancy in Planning:** The importance of establishing backup suppliers and diversifying funding sources to mitigate project risk.
- **Avoiding Over-Design:** Developed a better understanding of safety factors and how to design robust-yet-efficient structures without unnecessary mass.
- **Design for Manufacture:** How to design parts based on the actual constraints and capabilities of available manufacturing equipment.

Skills Demonstrated Team Leadership, FEA (ANSYS), CAD (SOLIDWORKS), Design for Manufacturing, Project Management, Technical Documentation, Composite Manufacturing, Systems Integration.

Table 2: Main Technical Specifications of Fulmair A.

Parameter	Figure
Wet Mass	30,980 g
Dry Mass	25,253 g
Predicted apogee	3,483m (OR)
Desired apogee	3,000m
Motor	Cesaroni 10367N1800-P
Propellant Mass	5,727 g
All-out Length	2900mm
Outer Diameter (OD)	160mm
Fins	4x
Drogue Descent Speed	39.92 m/s
Main Descent Speed	6 m/s
Active Flight Control	Yes, apogee control
Payload Form Factor	3U CubeSat
Payload Mass	4,000g

Figure 5: FULMAIR Project Gallery



3 Ashton Fire (Internship)

3.1 Project 1: Fire Strategy Development for Live Commercial & Residential Projects

Context As an intern at Ashton Fire, I contributed to the development of comprehensive fire safety strategies for three active construction projects in London, including a leisure center and a large-scale residential development. These strategies ensured compliance with UK building regulations and maximized occupant safety.

(Note: Unfortunately, due to a client NDA specific project details and visuals cannot be shared.)

My Contributions

- Authored detailed fire strategy reports outlining safety measures from construction to final occupancy.
- Applied national safety standards (Approved Document B Vol. 1 & 2, BS 9991, BS 9999) to inform design decisions.
- Collaborated with senior engineers to analyze building layouts and define requirements for fire resistance, smoke control, and means of egress.

Skills Demonstrated Technical Writing, Regulatory Compliance, Risk Assessment, Project Management, Stakeholder Communication.

3.2 Project 2: CFD Modeling for Occupant Egress

Context This project involved conducting a feasibility study to validate the safety of a proposed playground layout by modeling fire scenarios and calculating the time required for safe evacuation.

My Contributions

- Constructed the playground geometry in the CFD software B-Risk to create a high-fidelity model.
- Simulated various fire and ventilation conditions to determine worst-case scenarios for smoke spread and heat exposure.
- Calculated occupant egress times to ensure they met safety criteria under all simulated conditions.
- Performed a sensitivity analysis by varying model geometry to identify the limitations and critical parameters of the B-Risk solver.

Skills Demonstrated CFD (B-Risk), Computational Modeling, Egress Analysis, Sensitivity Analysis, Model Validation, Safety Engineering.

3.3 Project 3: Development of Engineering Analysis & Simulation Tools

Context To improve the efficiency of early-stage design analysis, I developed two custom calculation tools in Excel to perform rapid assessments of common fire engineering problems.

My Contributions

- **External Fire Spread Calculator:** Built an Excel-based solver to calculate the minimum required separation distance between buildings to prevent fire spread via radiative heat transfer. The model incorporates 3D view factors and the percentage of unprotected area on building facades.
- **Two-Zone Fire Model:** Created a simplified two-zone model to simulate fire growth in an enclosed space. The tool provides rapid, conservative estimates of when tenability conditions (temperature, plume layer height) are breached, saving significant time compared to full CFD runs.
- **Model Validation:** Conducted a comparative study between my simplified zone model and the high-fidelity B-Risk software to define the zone model's geometrical and solver limitations.

Skills Demonstrated Python, MS Excel, Heat Transfer.

3.4 Project 4: Research Paper on UK Fire Regulations & Demographic Change

Context I authored a paper analyzing how future demographic shifts in England will challenge existing fire safety regulations and guidance.

My Contributions

- Researched and synthesized current fire safety legislation and building codes in England.
- Analyzed national demographic data and projections to identify emerging risk factors (e.g., aging population, changes in disabilities and physical characteristics).
- Formulated recommendations for future-proofing fire safety guidance to accommodate societal changes.

Skills Demonstrated Research & Analysis, Regulatory Analysis, Strategic Thinking, Technical Writing.

4 Programming Projects

4.1 Dynamic Heat Pump & Building Thermal Model (Group Project)

Project Goal To develop a dynamic simulation tool in Python that models the performance of a heat pump and thermal storage system for various building types. The goal was to create a user-friendly interface to analyze system efficiency, optimize parameters, and understand the interplay between the building, environment, and heating system.

Links  [Link to GitHub Repository](#)

My Role & Core Contributions As a lead developer on this group project, I was responsible for writing the complete Python code for the simulation engine and the graphical user interface (GUI).

- **Developed the Simulation Engine in Python:**
 - Built the core model from first principles, implementing the differential equations governing heat transfer for the thermal storage tank.
 - Used the SciPy library's odeint solver to numerically solve the system's state over a 24-hour period.
 - Integrated the meteostat library to pull real-world, date-specific weather data for Edinburgh, ensuring the model was grounded in realistic environmental conditions.
- **Designed and Implemented a Graphical User Interface (GUI):**
 - Used the Tkinter library to create an intuitive interface allowing users to select pre-set building types (e.g., Factory, apartment) or input custom system parameters.
 - The UI provides real-time visualization of results, plotting key metrics like tank temperature over time and final performance values using Matplotlib.
- **Engineered the System Control Logic:**
 - Designed and implemented a hysteresis control loop to manage the heat pump's ON/OFF state. This prevented rapid, unstable switching that would occur from a simple threshold check, demonstrating an understanding of practical control systems.
- **Conducted Performance Analysis & Optimization:**
 - The model calculated key performance metrics, including the Energy Consumption Ratio and efficiency relative to a Carnot cycle.
 - Used the simulation tool to run an optimization study, adjusting parameters like U-values, set-points, and thermal mass to successfully reduce the system's total energy expenditure while still meeting heating demands.

Skills Demonstrated

- **Programming Software:** Python, Tkinter (GUI Development), SciPy, NumPy, Matplotlib
- **Engineering Analysis:** Systems-Level Modeling, Thermal Dynamics Heat Transfer, Numerical Methods (ODE Solvers), Control Systems (Hysteresis), Data Analysis Visualization, System Optimization
- **Professional Skills:** Technical Writing Documentation, User Interface (UI) Design, Collaborative Project Management

4.2 Fire Engineering Toolkit

Context During my internship, I often used Excel for quick-fire engineering calculations. I thought it would be a great learning experience to build a more robust version of those tools in Python. I developed this standalone desktop app to practice my programming and create something useful for the kind of analysis I was doing.

 [Link to GitHub Repository](#)

What I Built

- **Simple User Interface:** To make the tools easy to use, I built a graphical interface using the CustomTkinter library. It allows anyone to enter parameters and run calculations without needing to see the code.
- **Two-Zone Fire Model:** I wrote the code for a simplified fire model using standard fire engineering equations. It predicts how much time it takes for conditions to become untenable.
- **External Fire Spread Calculator:** I added a second tool that calculates how far apart buildings need to be to prevent fire from spreading through radiative heat transfer. It considers the size of the unprotected area, distance from the boundary, and whether sprinklers are present.
- **Data Export to Excel:** I included a feature to save multiple calculation results. With one click, it can export all the data to a clean Excel spreadsheet.

Skills Demonstrated

- **Programming:** Python, CustomTkinter (GUI)
- **Data Libraries:** NumPy, Pandas, Matplotlib
- **Engineering Concepts:** Fire Dynamics, Heat Transfer, Two-Zone Modeling
- **Practical Skills:** UI Design, Data Management, Problem-Solving

5 Personal Projects

5.1 Knifemaking

Context As a teen, I was completely hooked on the YouTube channel "Man at Arms," watching blacksmiths bring fictional and historical weapons to life. This sparked a years-long hobby where I built a small workbench and dove headfirst into bladesmithing. It became my practical lab for understanding materials science—things like heat treatment, and alloys which proved extremely useful years later in university.

- **Learned & Designed:** Studied the history and function of different blade designs to create my own functional pieces.
- **Built from Scratch:** Took steel bars, sheets, and a lot of street mystery metal and shaped them using a combination of power tools (angle grinder, drill press) and hand tools for precision work.
- **Attempted to master heat treatment:** Applied core materials science concepts to heat treat my blades, carefully controlling temperature and time to normalize, quench, and temper the steel for optimal hardness and toughness. Using a BBQ for heat, my grandma's 30 year old hair dryer for air supply, and, in a moment of stupidity, extra virgin olive oil from my grandparents olive trees for the quench. Looking back, it was a waste of good oil, but it was also a fantastic lesson in material properties that you just can't get from a textbook.
- **Finished with Detail:** Crafted custom wooden handles, performed multi-stage polishing, and used acid etching to bring out the unique patterns in the steel.

Skills Demonstrated Metallurgy, Heat Treatment, Steel cold-working, Hands-on Fabrication, Metal & Woodworking, Design, Problem-Solving, Workshop Safety, Self-Motivation, Use of Power & Hand Tools.



Figure 7: From left to right: Axe, Seax, Bowie, Repurposed butter knife, Spoon



Figure 6: Nepalese Kukri, and karambit (1st and 2nd attempt)

5.2 The BBQ Project

The BBQ was an interesting and extremely fun project done with my flatmate. There is an extensive explanation on how it was made why and the mistakes made in the

following article written by my flatmate:
[The Stainless Steel BBQ Project on Notion](#)